

SQL Query

AGGREGATE FUNCTIONS (15 Questions)

1. Find highest salary.

```
mysql> select emp_id,emp_name,salary from employees
-> order by salary desc
-> limit 1;
+-----+-----+-----+
| emp_id | emp_name | salary |
+-----+-----+-----+
|      5 | Rohan    | 90000.00 |
+-----+-----+-----+
```

2. Find second highest salary.

```
mysql> select emp_id,emp_name,salary from employees
-> order by salary desc
-> limit 1 offset 1;
+-----+-----+-----+
| emp_id | emp_name | salary |
+-----+-----+-----+
|     14 | Mohit    | 87000.00 |
+-----+-----+-----+
1 row in set (0.011 sec)
```

3. Find lowest salary.

```
mysql> select emp_id,emp_name,salary from employees
-> order by salary asc
-> limit 1;
+-----+-----+-----+
| emp_id | emp_name | salary |
+-----+-----+-----+
|     10 | Kiran    | 36000.00 |
+-----+-----+-----+
1 row in set (0.009 sec)
```

4. Find average salary.

```
mysql> select avg(salary) as Average_Salary
-> from employees;
+-----+
| Average_Salary |
+-----+
| 58933.333333 |
+-----+
1 row in set (0.080 sec)
```

```
mysql> select d.dept_name,d.dept_id,avg(salary) as Average_Salary from employees as e
-> join departments as d
-> on e.dept_id = d.dept_id
-> group by d.dept_name,d.dept_id;
+-----+-----+-----+
| dept_name | dept_id | Average_Salary |
+-----+-----+-----+
| HR        | 101     | 42333.333333 |
| IT        | 102     | 61750.000000 |
| Finance   | 103     | 54333.333333 |
| Sales     | 104     | 77333.333333 |
| Marketing | 105     | 57500.000000 |
+-----+-----+-----+
5 rows in set (0.039 sec)
```

5. Find total salary.

```
mysql> select sum(salary) as Total_Salary from employees;
+-----+
| Total_Salary |
+-----+
|      884000.00 |
+-----+
1 row in set (0.024 sec)
```

```
mysql> select d.dept_id,d.dept_name,sum(salary) as Total_Salary from employees as e
-> join departments d
-> on e.dept_id = d.dept_id
-> group by d.dept_id,d.dept_name;
+-----+-----+-----+
| dept_id | dept_name | Total_Salary |
+-----+-----+-----+
|      101 | HR        |      127000.00 |
|      102 | IT        |      247000.00 |
|      103 | Finance   |      163000.00 |
|      104 | Sales     |      232000.00 |
|      105 | Marketing |      115000.00 |
+-----+-----+-----+
5 rows in set (0.051 sec)
```

6. Count employees.

```
mysql> select d.dept_id,d.dept_name,count(*) as Total_Employee from employees as e
-> join departments d
-> on e.dept_id = d.dept_id
-> group by d.dept_id,d.dept_name;
+-----+-----+-----+
| dept_id | dept_name | Total_Employee |
+-----+-----+-----+
|      101 | HR        |              3 |
|      102 | IT        |              4 |
|      103 | Finance   |              3 |
|      104 | Sales     |              3 |
|      105 | Marketing |              2 |
+-----+-----+-----+
5 rows in set (0.061 sec)
```

```
mysql> select count(*) as Total_Employee from employees
-> ;
+-----+
| Total_Employee |
+-----+
|              15 |
+-----+
1 row in set (0.050 sec)
```

7. Count female employees.

```
mysql> SELECT count(*) as Female_Employees from employees
-> where gender='Female';
+-----+
| Female_Employees |
+-----+
|              8 |
+-----+
1 row in set (0.008 sec)
```

```
mysql> SELECT gender, count(*) as Total_Employees from employees
-> where gender = 'Female';
```

gender	Total_Employees
Female	8

```
1 row in set (0.008 sec)
```

```
mysql> SELECT gender, COUNT(*) AS Total_Employees
-> FROM employees
-> GROUP BY gender;
```

gender	Total_Employees
Male	7
Female	8

```
2 rows in set (0.010 sec)
```

8. Count distinct departments.

```
mysql> SELECT COUNT(DISTINCT dept_id) AS Total_Departments
-> FROM employees;
```

Total_Departments
5

```
1 row in set (0.008 sec)
```

9. Find total project hours.

```
mysql> SELECT project_name,
-> TIMESTAMPDIFF(HOUR, start_date, end_date) AS project_hours
-> FROM projects;
```

project_name	project_hours
AI Chatbot	8736
ERP System	13872
Website	4776
Mobile App	11472
CRM System	6072

```
mysql> SELECT DATEDIFF(end_date, start_date) AS project_days
-> FROM projects;
```

project_days
364
578
199
478
253

```
5 rows in set (0.093 sec)
```

```
mysql> SELECT TIMESTAMPDIFF(MONTH, start_date, end_date) AS project_months
-> FROM projects;
```

project_months
11
18
6
15
8

```
5 rows in set (0.022 sec)
```

```
mysql> ALTER TABLE projects
-> ADD project_hours INT;
Query OK, 0 rows affected (1.031 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> UPDATE projects
-> SET project_hours = 200
-> WHERE project_id = 201;
Query OK, 1 row affected (0.103 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from projects;
+-----+-----+-----+-----+-----+-----+
| project_id | project_name | start_date | end_date | budget | project_hours |
+-----+-----+-----+-----+-----+-----+
| 201 | AI Chatbot | 2023-01-01 | 2023-12-31 | 1000000.00 | 200 |
| 202 | ERP System | 2022-05-01 | 2023-11-30 | 800000.00 | 440 |
| 203 | Website | 2023-03-15 | 2023-09-30 | 400000.00 | NULL |
| 204 | Mobile App | 2022-10-10 | 2024-01-31 | 1200000.00 | 744 |
| 205 | CRM System | 2023-02-20 | 2023-10-31 | 600000.00 | 700 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.008 sec)
```

10.Find maximum project budget.

```
mysql> select max(budget) as Max_Budget from projects;
+-----+
| Max_Budget |
+-----+
| 1200000.00 |
+-----+
1 row in set (0.009 sec)
```

11.Find minimum project budget.

```
mysql> select min(budget) as Min_Budget from projects;
+-----+
| Min_Budget |
+-----+
| 400000.00 |
+-----+
1 row in set (0.026 sec)
```

12.Find average employee age.

```
mysql> select avg(budget) as Average_Budget from projects;
+-----+
| Average_Budget |
+-----+
| 800000.000000 |
+-----+
1 row in set (0.007 sec)
```

13.Find total number of projects.

```
mysql> select sum(budget) as Total_Budget from projects;
+-----+
| Total_Budget |
+-----+
| 4000000.00 |
+-----+
1 row in set (0.008 sec)
```

14.Find average hours worked.

```
mysql> select avg(project_hours) as Average_Hour from projects;
+-----+
| Average_Hour |
+-----+
|      521.0000 |
+-----+
1 row in set (0.008 sec)

mysql> SELECT AVG(TIMESTAMPDIFF(HOUR, start_date, end_date)) AS Average_Project_Hours
-> FROM projects;
+-----+
| Average_Project_Hours |
+-----+
|          8985.6000 |
+-----+
1 row in set (0.012 sec)
```

15.Find total employees hired after 2021.

```
mysql> SELECT COUNT(*) AS Total_Employees
-> FROM employees
-> WHERE joining_date >= '2022-01-20';
+-----+
| Total_Employees |
+-----+
|          5 |
+-----+
1 row in set (0.043 sec)
```

```
mysql> select * from employees;
+-----+-----+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | gender | age | salary | joining_date | dept_id | manager_id |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Amit | Male | 28 | 45000.00 | 2021-02-15 | 102 | NULL |
| 2 | Neha | Female | 25 | 38000.00 | 2022-01-20 | 101 | 1 |
| 3 | Rahul | Male | 32 | 70000.00 | 2020-08-18 | 102 | 1 |
| 4 | Priya | Female | 29 | 65000.00 | 2019-05-10 | 103 | 3 |
| 5 | Rohan | Male | 35 | 90000.00 | 2018-03-22 | 104 | NULL |
| 6 | Sneha | Female | 27 | 42000.00 | 2023-01-12 | 101 | 2 |
| 7 | Arjun | Male | 31 | 80000.00 | 2017-06-05 | 102 | 3 |
| 8 | Pooja | Female | 26 | 39000.00 | 2022-10-10 | 105 | 5 |
| 9 | Vikas | Male | 30 | 55000.00 | 2021-07-01 | 104 | 5 |
| 10 | Kiran | Female | 24 | 36000.00 | 2023-04-14 | 103 | 4 |
| 11 | Ankit | Male | 29 | 52000.00 | 2021-12-11 | 102 | 3 |
| 12 | Meera | Female | 33 | 76000.00 | 2019-09-21 | 105 | 5 |
| 13 | Riya | Female | 28 | 47000.00 | 2022-06-16 | 101 | 2 |
| 14 | Mohit | Male | 34 | 87000.00 | 2018-11-19 | 104 | 5 |
| 15 | Nisha | Female | 31 | 62000.00 | 2020-01-09 | 103 | 4 |
+-----+-----+-----+-----+-----+-----+-----+-----+
15 rows in set (0.023 sec)
```